Immediate Recall, Later Word Recognition, and Information Congruency in Reading and Listening Comprehension

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Background: Recent studies [1,2] have documented systematic differences in the extent to what readers recall certain types of information immediately after reading a sentence. For example, information conveyed by direct objects tends to be recalled significantly better than information conveyed by temporal or locative adjuncts. This can be interpreted as a selective attention process [2]. The present study examines immediate recall of information conveyed by agents and locative adjuncts (Loc) in Czech and how it is influenced by adjunct congruency (or contextual fit) [3,4]. Moreover, in two of our experiments, later word recognition is tested [4].

Method: First, predictability of Loc in combination with 57 transitive verbs was normed (N=115 Czech speakers). 24 of these combinations were then used for creating stimuli for further experiments. Four reading experiments were conducted using a self-paced reading paradigm with whole sentences appearing at once for experiments 1 (N=195) and 5 (N=240) and with sentences presented word-by-word for experiments 2 (N=183) and 6 (N=182). Once the sentence disappeared, an open-ended question was shown targeting either the agent (Who did it?), or the Loc (Where did it happen?). Additionally, experiments 5 and 6 were followed by a later word recognition task including a set of 48 agents and 48 Loc, half of which appeared in the experimental sentences and half was absent from them. Then, two listening experiments were conducted. Experiment 3 (N=208) used stimuli audio-recorded by native speakers of Czech with flat intonation. For experiment 4 (N=213), stimuli generated by an artificial intelligence were used. The open-ended questions were visually presented, and participants responded by typing. All experiments use the same 24 experimental items and 72 fillers and manipulate word order, information targeted by the comprehension question and Loc congruency (see Table 1).

Result: Fig. 1 shows the differences in recall accuracy between the conditions in all experiments. The nested logit mixed-effects model showed a general recall difference for listening experiments (but not for the reading ones): agents were recalled better than Loc. Moreover, the model yielded a significant effect of congruency for Loc recall in all experiments and for agent recall in experiments 1 and 6. Effect of position for agents was found in experiments 2 and 6: better recognition was observed for medial position. Results of the later word recognition task are illustrated in Fig. 2. A congruency effect was observed only for words that did not appear in the reading task. A question effect was found for both words from experimental items (Loc recognized more accurately) and words absent from the stimuli (agents recognized more accurately). In general, participants demonstrated better recognition of words absent from the experimental stimuli compared to words present in them.

Discussion: Overall, our experiments demonstrated a systematic advantage of congruent Loc in the immediate recall task. However, no congruency effect was found in the later word recognition task, which we attribute to a weaker representation of the sentence context in memory due to the temporal distance. Moreover, in line with prior findings on Czech reading data [1,2], experiments involving spoken materials showed a tendency for better immediate recall of core information in sentences compared to accessory information. However, this effect was not replicated in reading, which points to differences in presentation modality.

Word order	Congruency	Sentence
ltvso	yes	V obchodě v neděli koupila Klára hrozně hezký pruhovaný tričko.
ltvso	no	V parku v neděli koupila Klára hrozně hezký pruhovaný tričko.
stvlo	yes	Klára v neděli koupila v obchodě hrozně hezký pruhovaný tričko.
stvlo	no	Klára v neděli koupila v parku hrozně hezký pruhovaný tričko.

Table 1: Item example. Word order values: ltvso = locative adjunct - temporal adjunct - verb - subject - object; stvlo = subject - temporal adjunct - verb - locative adjunct - object. Sentences have the same meaning and only differ in their word order and locative adjunct congruency: "Klára bought a really nice striped T-shirt in the store/in the park on Sunday."

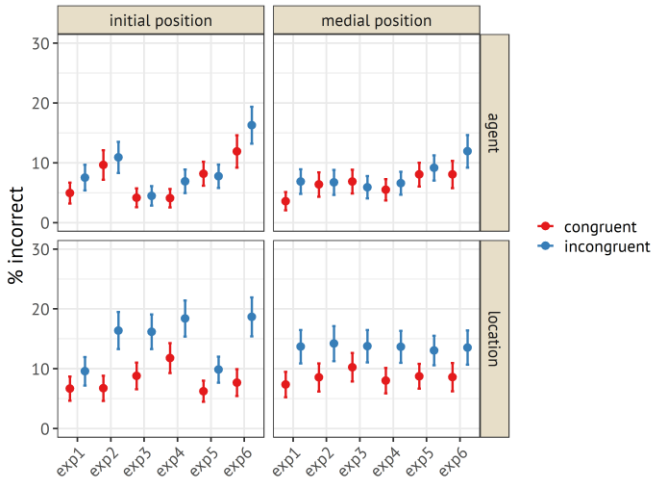

Figure 1: % of incorrect answers in all experiments. Congruent = congruent locative adjunct, incongruent = incongruent locative adjunct, exp1 = reading experiment, sentence presented at once, exp2 = reading experiment, sentence presented word-by-word, exp3 = listening experiment, stimuli recorded by real speakers, exp4 = listening experiment, stimuli generated by AI, exp5 = reading experiment, sentence presented at once, exp6 = reading experiment, sentence presented word-by-word.

Figure 2: % of incorrect answers in the later word recognition task in exp5 (reading experiment, sentence presented at once) and exp 6 (reading experiment, sentence presented word-by-word).

References

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